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Polish and Finalize Mechanism
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Born from the boundaries of affordance and accessibility, Braille's Bells is a doorbell system that celebrates the tactile experience from the carefully crafted handle to the textured buttons. Although at its core, our machine functions as a doorbell, its purpose is to cater to the blind. Through the tactile touch points of the handle and braille buttons, our machine grants the user autonomy to choose their experience. After the machine is initiated by the lever, the user can select a button and decide which bell to ring. The right button, marked with "one" in Braille, rings the bell once every rotation. Conversely, the left button, marked with "two" in Braille, rings the bell twice every rotation.

As an escapement mechanism controls the release of power from a falling weight, a sprocket transfers the rotary motion through a chain to another sprocket that transfers the same rotary motion onto a series of gears. Each gear rotates a sliding bevel gear. These bevel gears make up the clutch mechanism that is activated by the buttons on the front of the machine. As a button is pressed, the corresponding clutch mechanism is engaged, pressing the rotating bevel gear onto a stationary one at a 90° angle. This clutch mechanism transforms linear motion into rotary motion. As these gears catch, a cam and follower that is attached to the rotating bevel gear transforms the rotary motion into reciprocating motion. Finally, as the follower goes up and down, it pulls an oscillating hammer which rings the bell as the follower falls, translating reciprocating motion to oscillating motion.

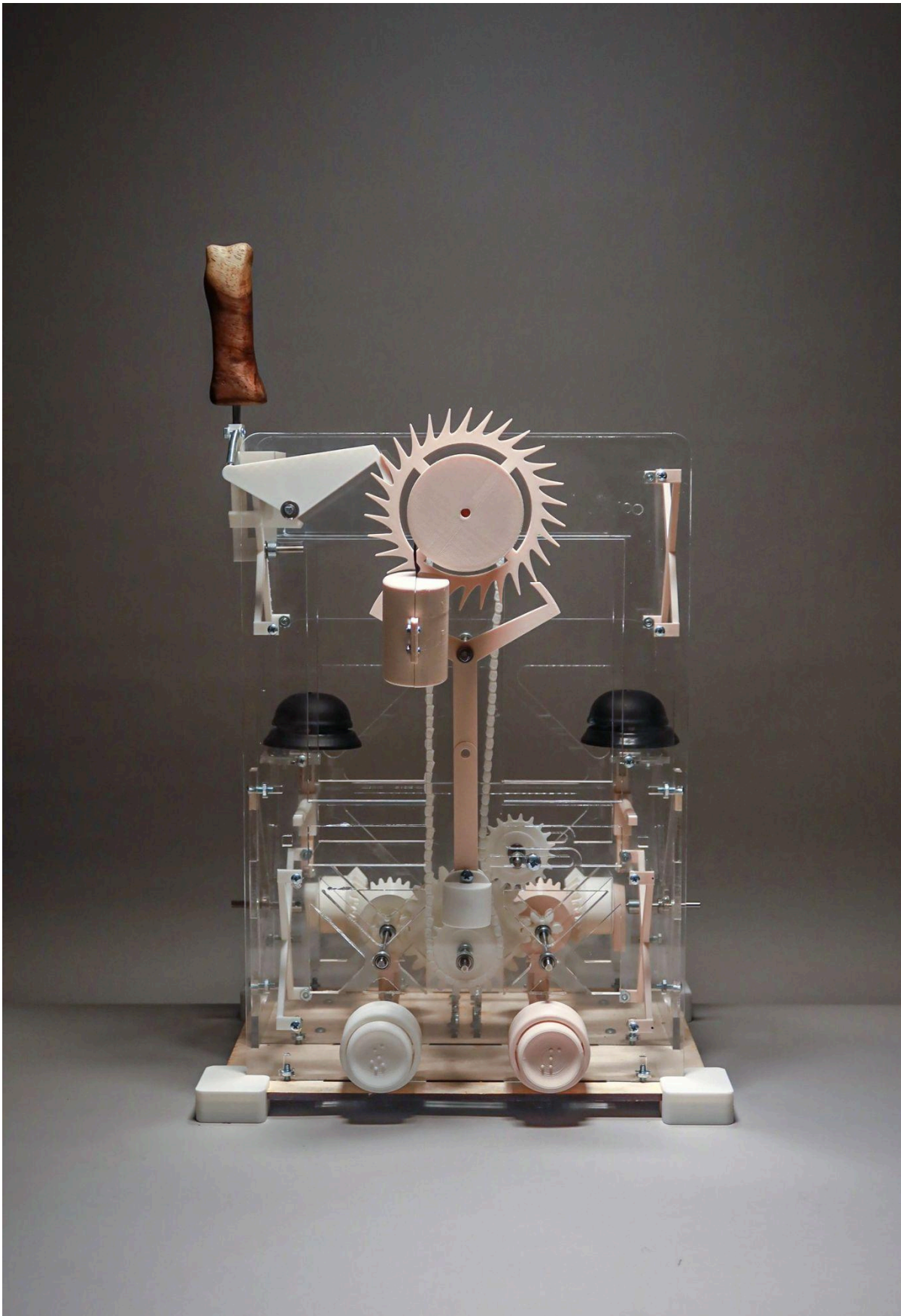


Figure 1: "Braille's Bells"

Close Ups:



Figure 2: Close-up of wooden lever actuator.

When creating the actuator, we naturally gravitated toward a lever-pulling motion. Consequently, through our shaped wooden handle, we hoped to create an affordance that would visually cue the user to pull it like a lever to start the machine.



Figure 3: Close-up of escapement mechanism and actuator.

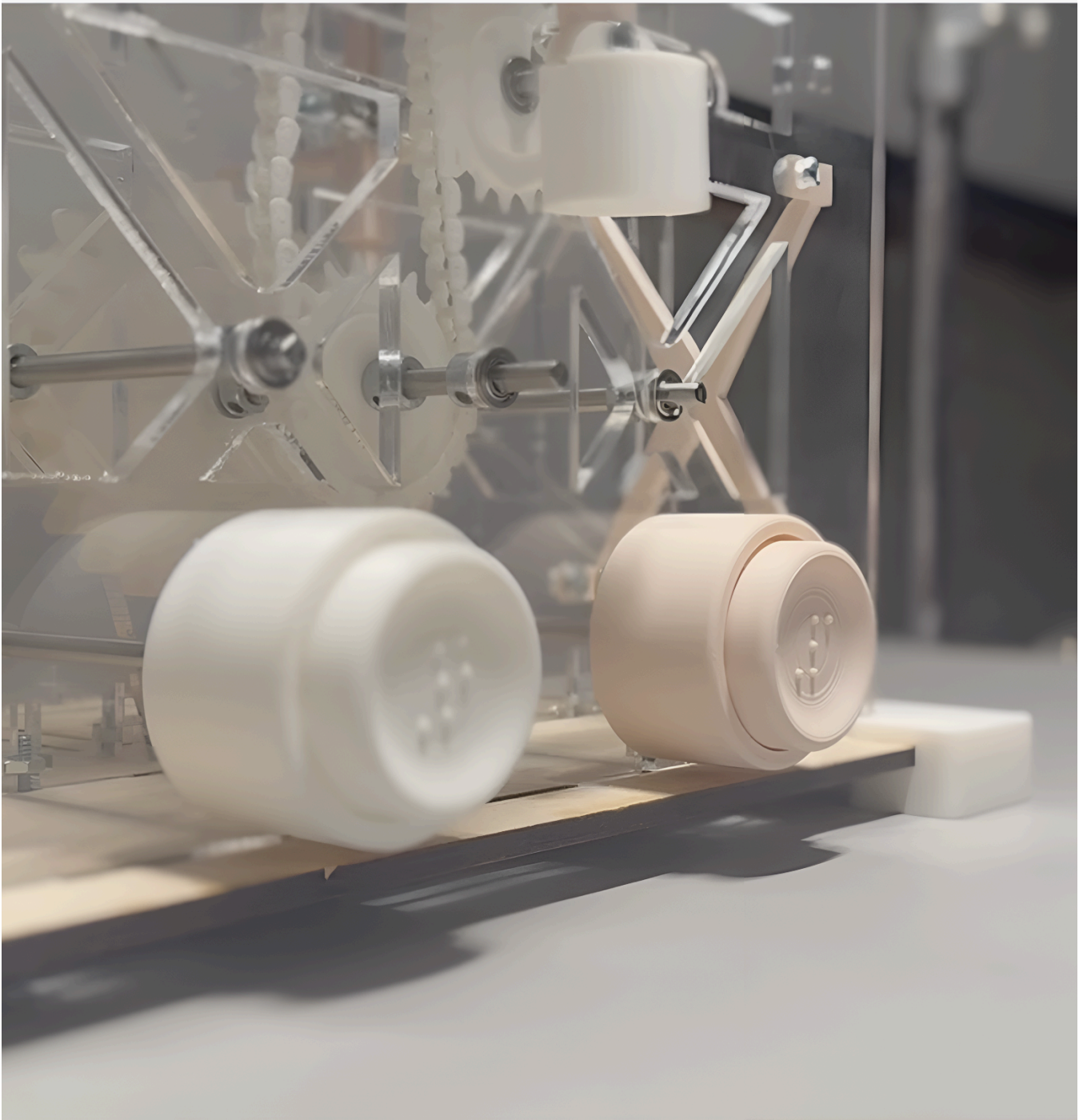


Figure 4: Close-up of buttons with braille writing.

We wanted our user to have a unique and autonomous experience. Given our machine caters to the blind, the right button, marked with “one” in Braille, rings the bell once every rotation when pressed. Conversely, the left button, marked with “two” in Braille, rings the bell twice every rotation.

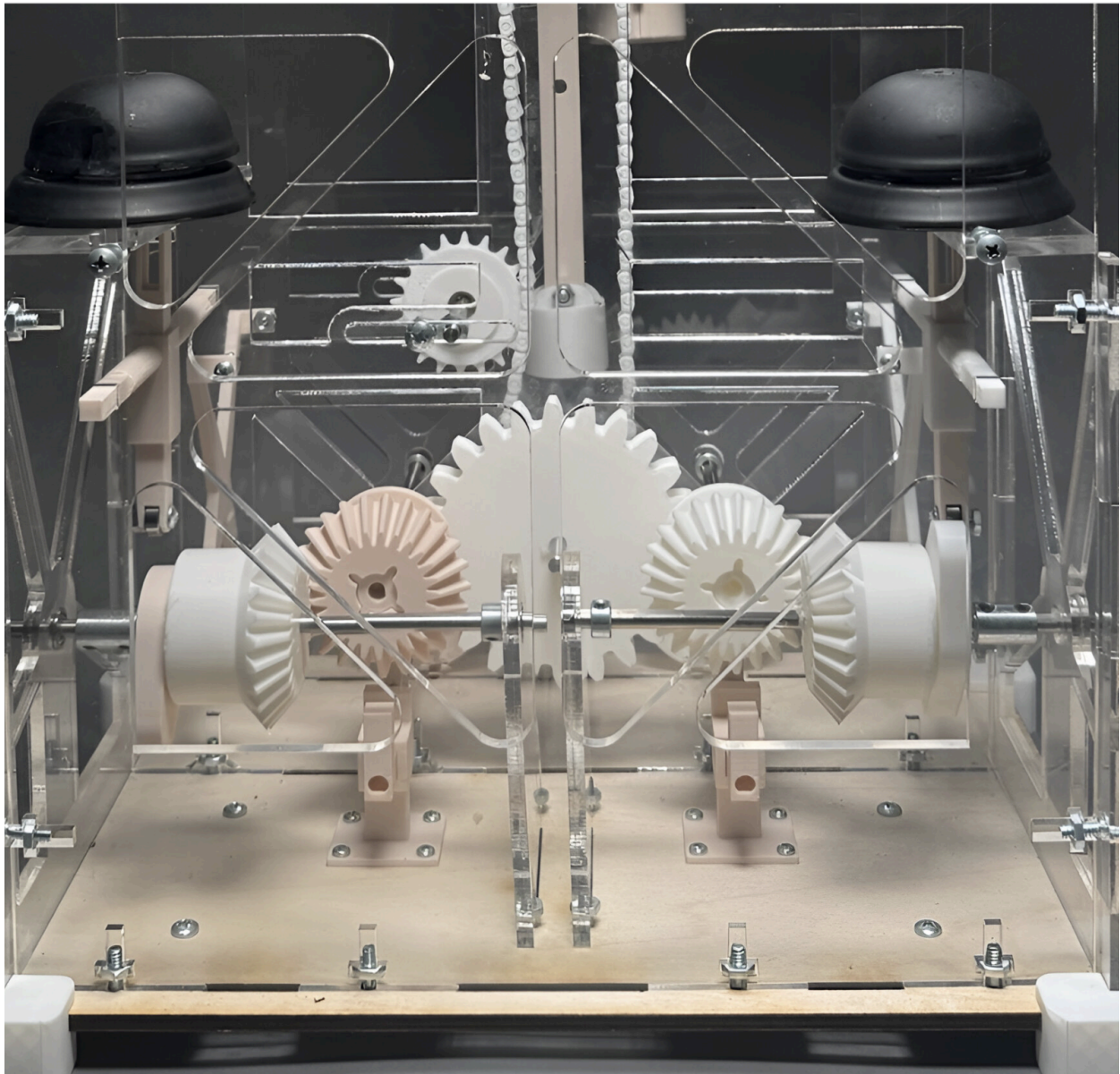


Figure 5: Rear-view of dual cam and followers.

On the left, the cam is one-sided, resulting in the follower ringing the bell once every rotation. Conversely, the cam on the right is two-sided, resulting in the follower ringing the bell twice every rotation.

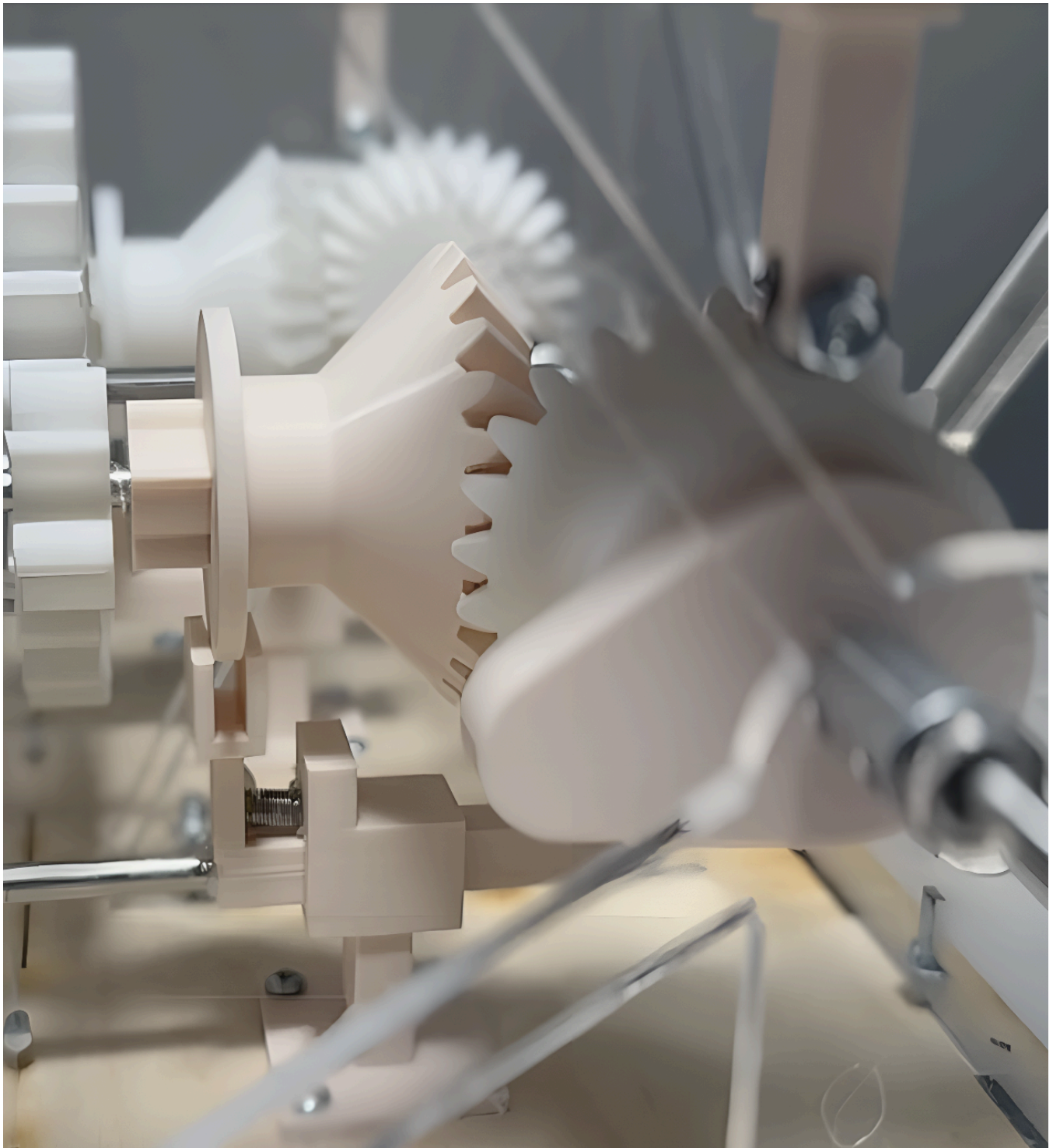


Figure 6: Close-up of engaged clutch mechanism.

When a button is pressed, the shaft is connected to a clutch mechanism. This pressed shaft compresses a spring, moving a sliding and rotating bevel gear to engage with a stationary bevel gear (that rotates the attached cam) at a 90° angle. This transforms linear motion into rotational motion. Additionally, when the button is released, the spring returns the clutch mechanism to its natural state, disengaging from the stationary bevel gear.

Process:



Figure 7: Original Door Knocker Mechanism

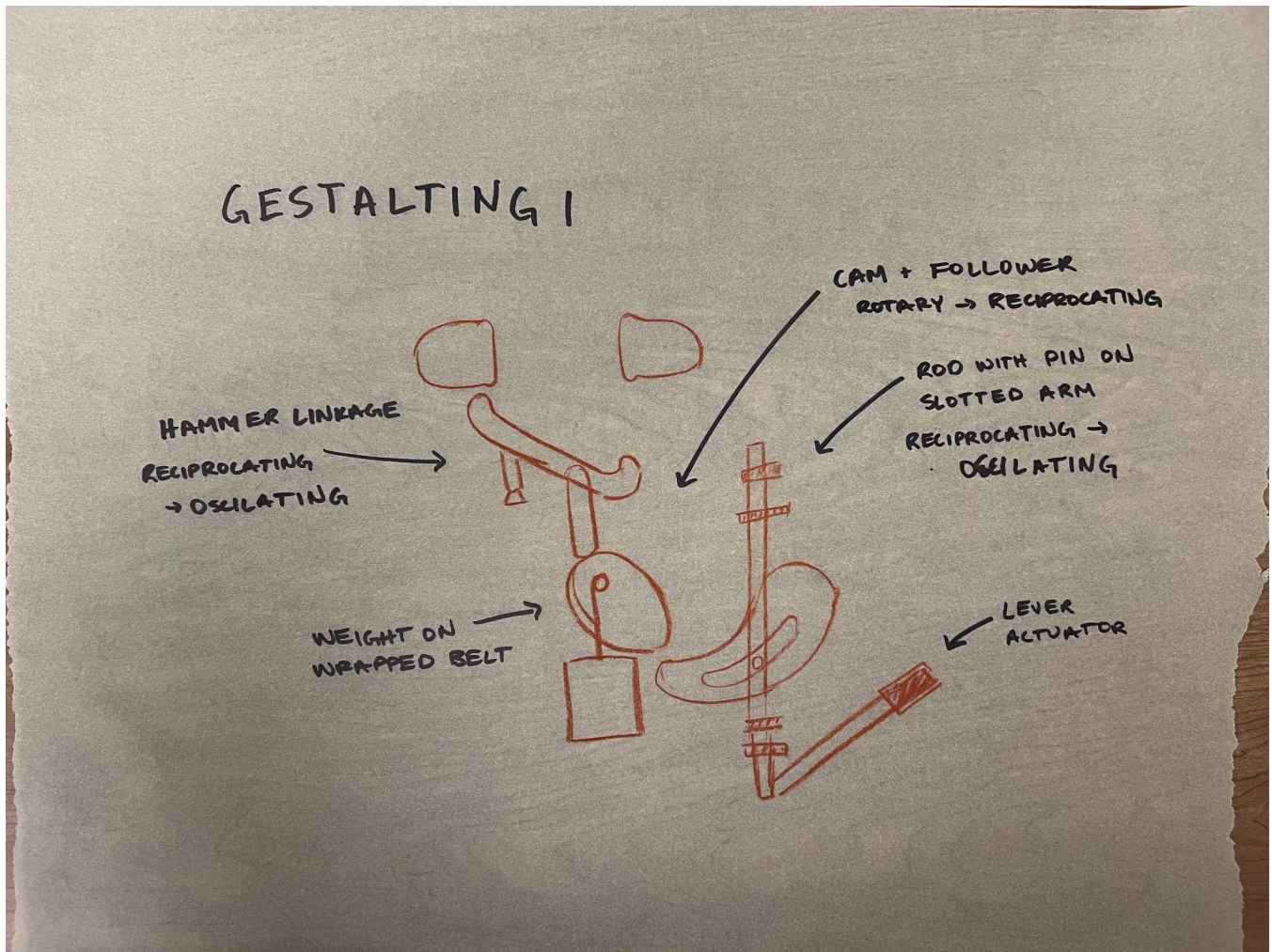


Figure 8: Gestalting stage from last MEDGI.

After three rounds of MEDGI chain, we transformed the original door knocker into a powered and actuated machine that would ring two bells.

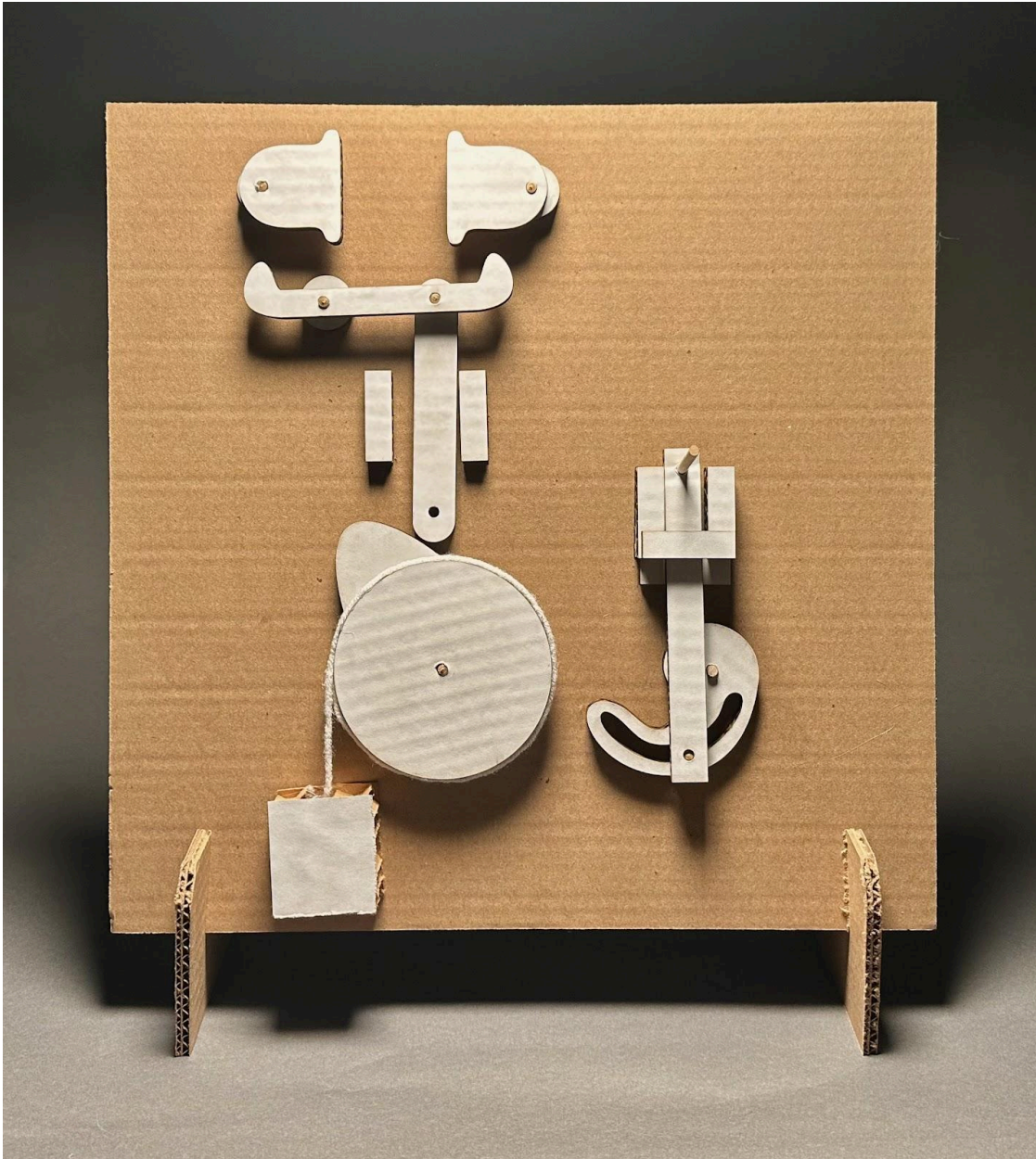


Figure 9: Cardboard prototype of our last MEDGI.

The design seen in Figure 8 was then represented in a cardboard model as a proof of concept.

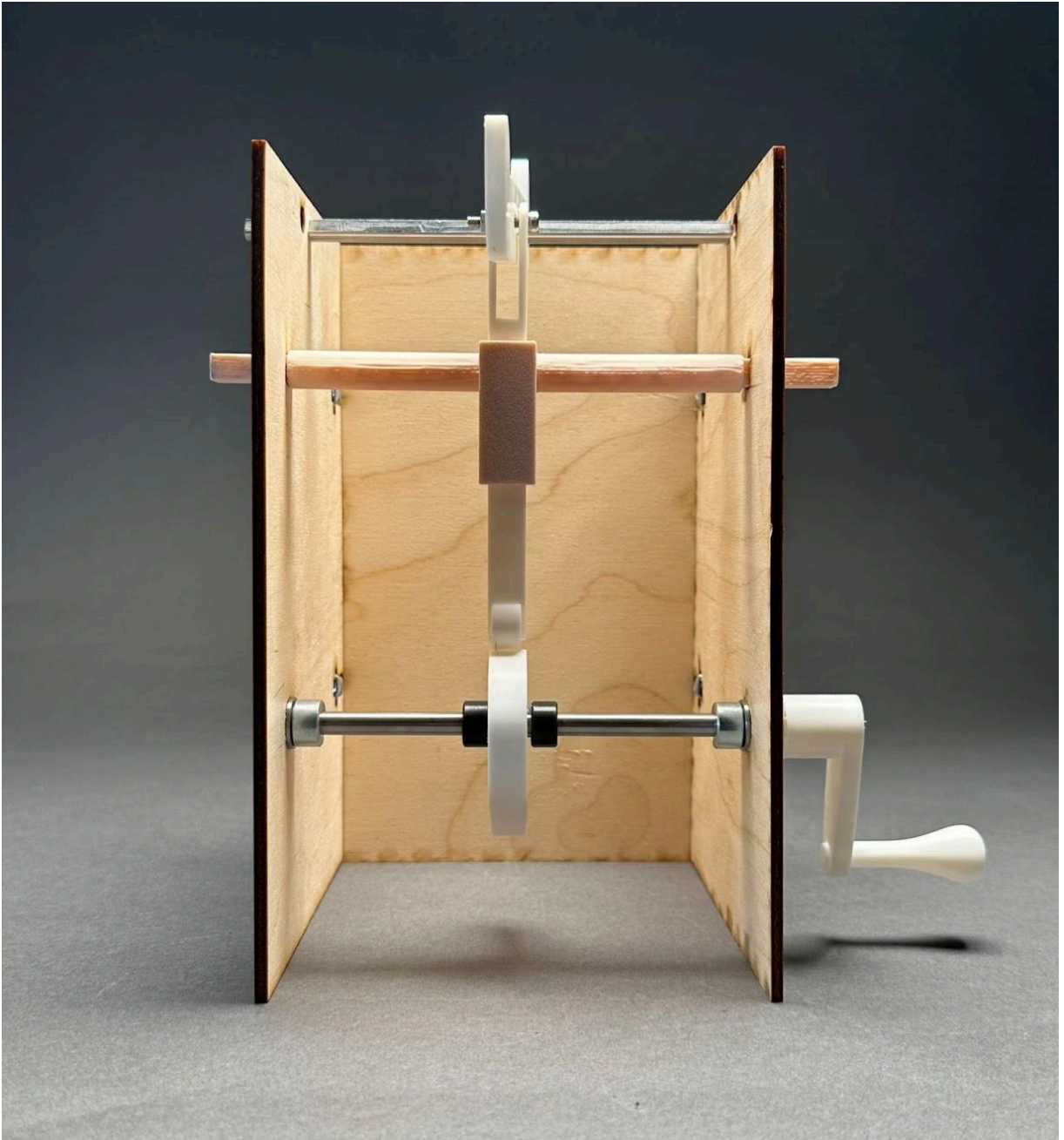


Figure 10: 3-dimensional representation of cam and follower critical mechanism.

From our cardboard model, we chose the cam and follower to be our critical mechanism. We then realized this critical mechanism in three dimensions.



Figure 11: First working iteration of full machine.

This figure demonstrates our first iteration of our machine. The escapement mechanism, powered by the falling weight, drives a sprocket and chain horizontally. This chain then drives another sprocket, which rotates the cam and follower and rings the bell.

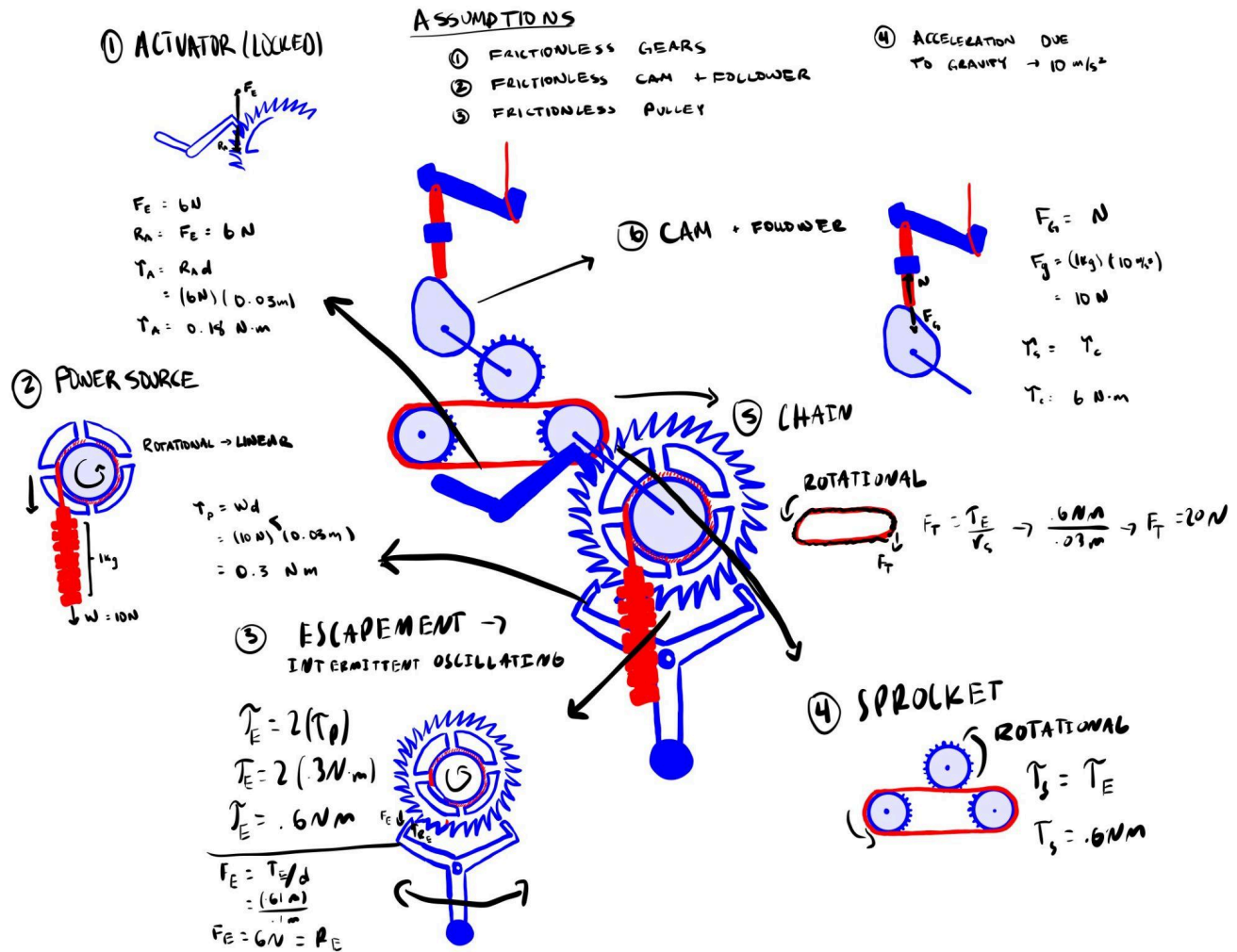


Figure 9: Force diagrams and calculations for the initial prototype.

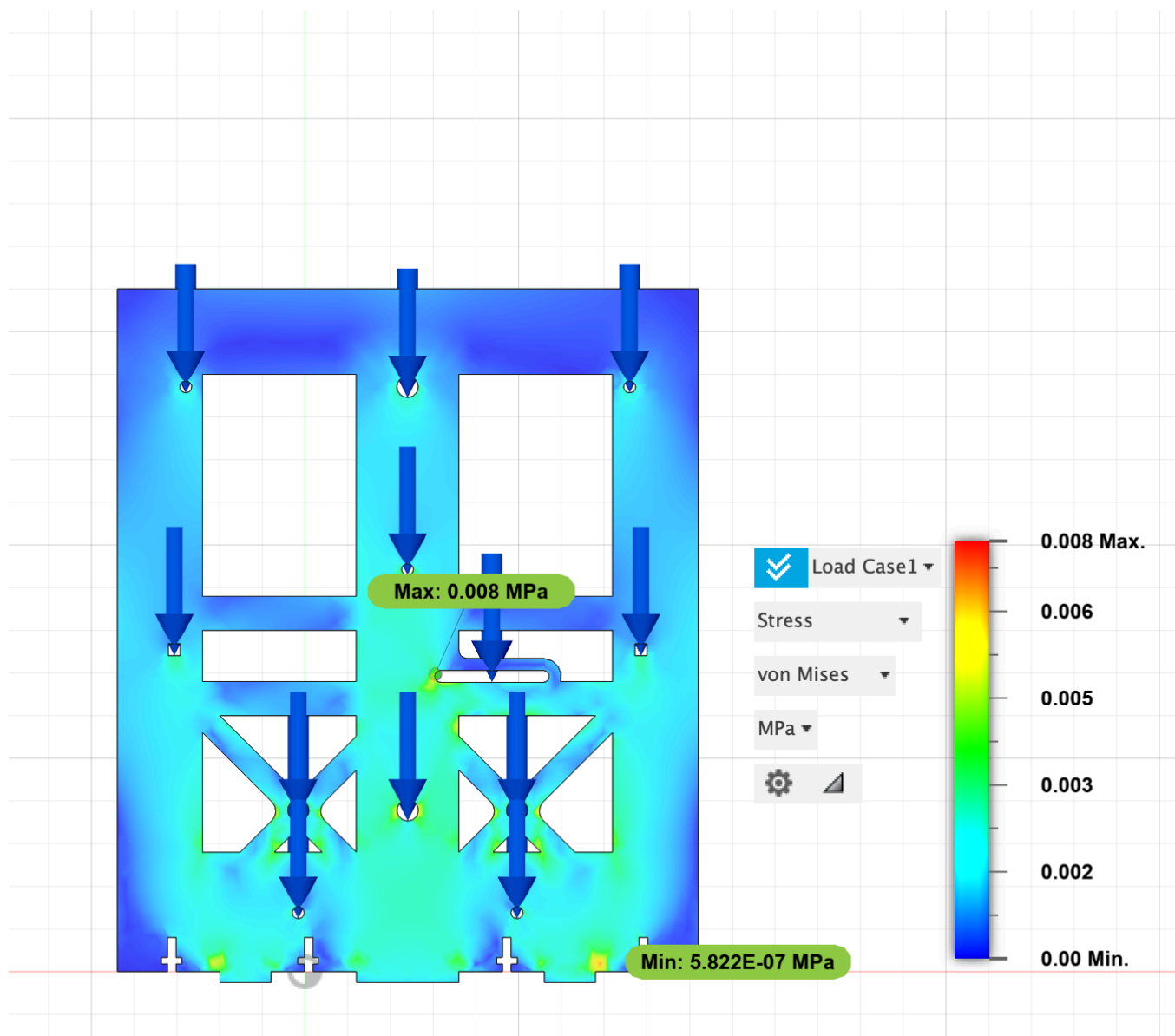


Figure 10: FEA and redesign of the front-most part of the frame.

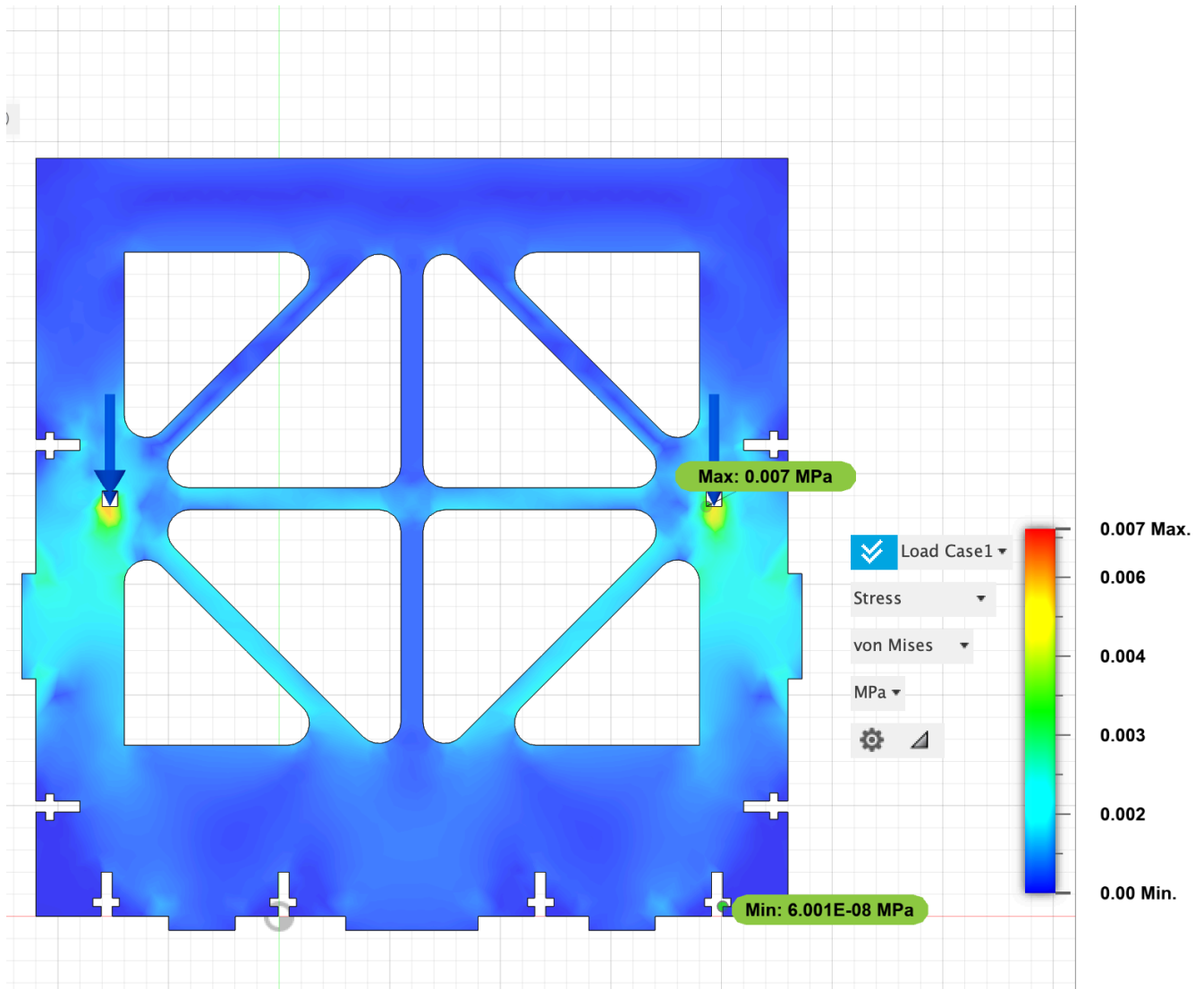


Figure 11: FEA and redesign of back-most frame.

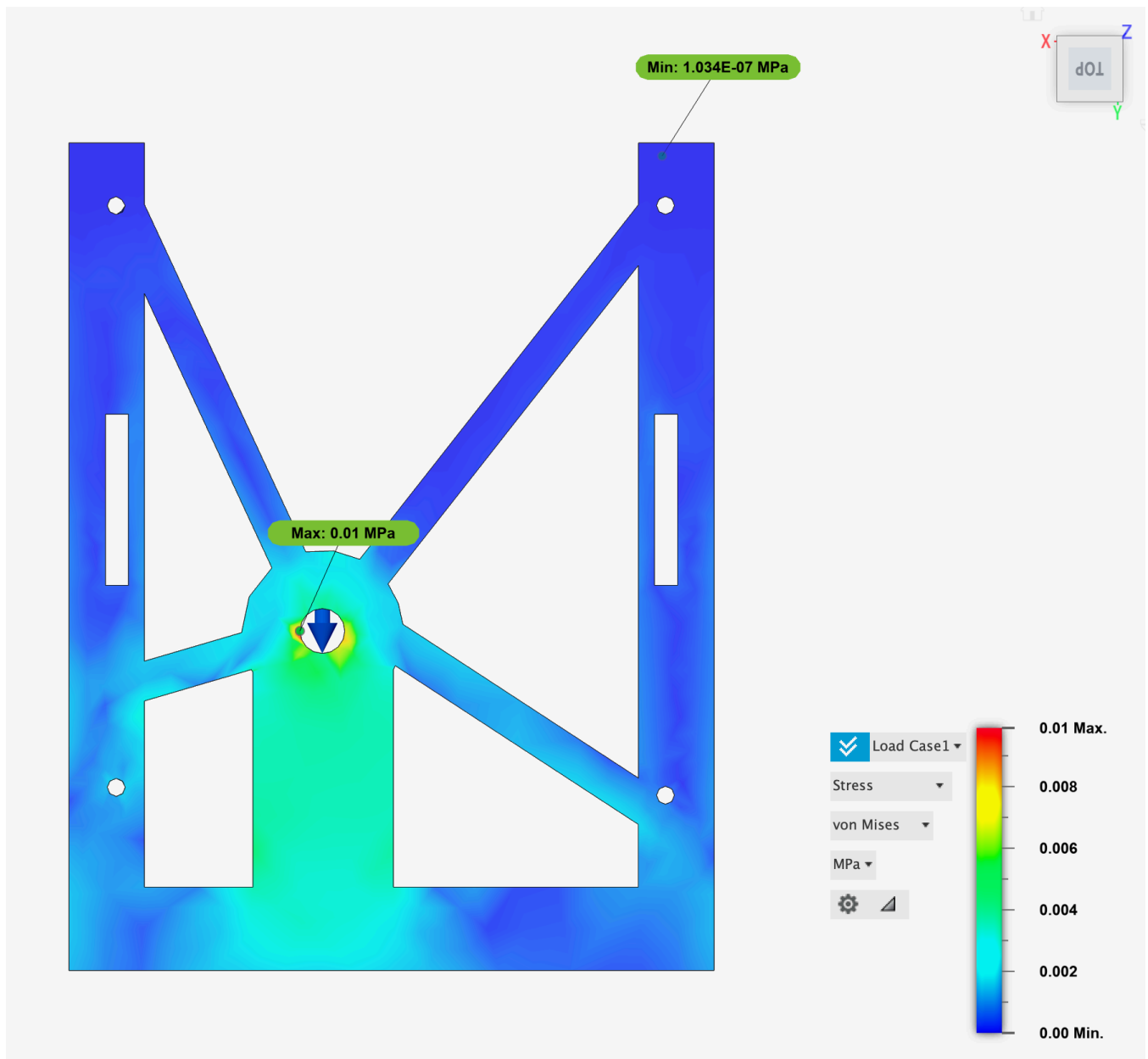


Figure 12: FEA and redesign of side of frame.

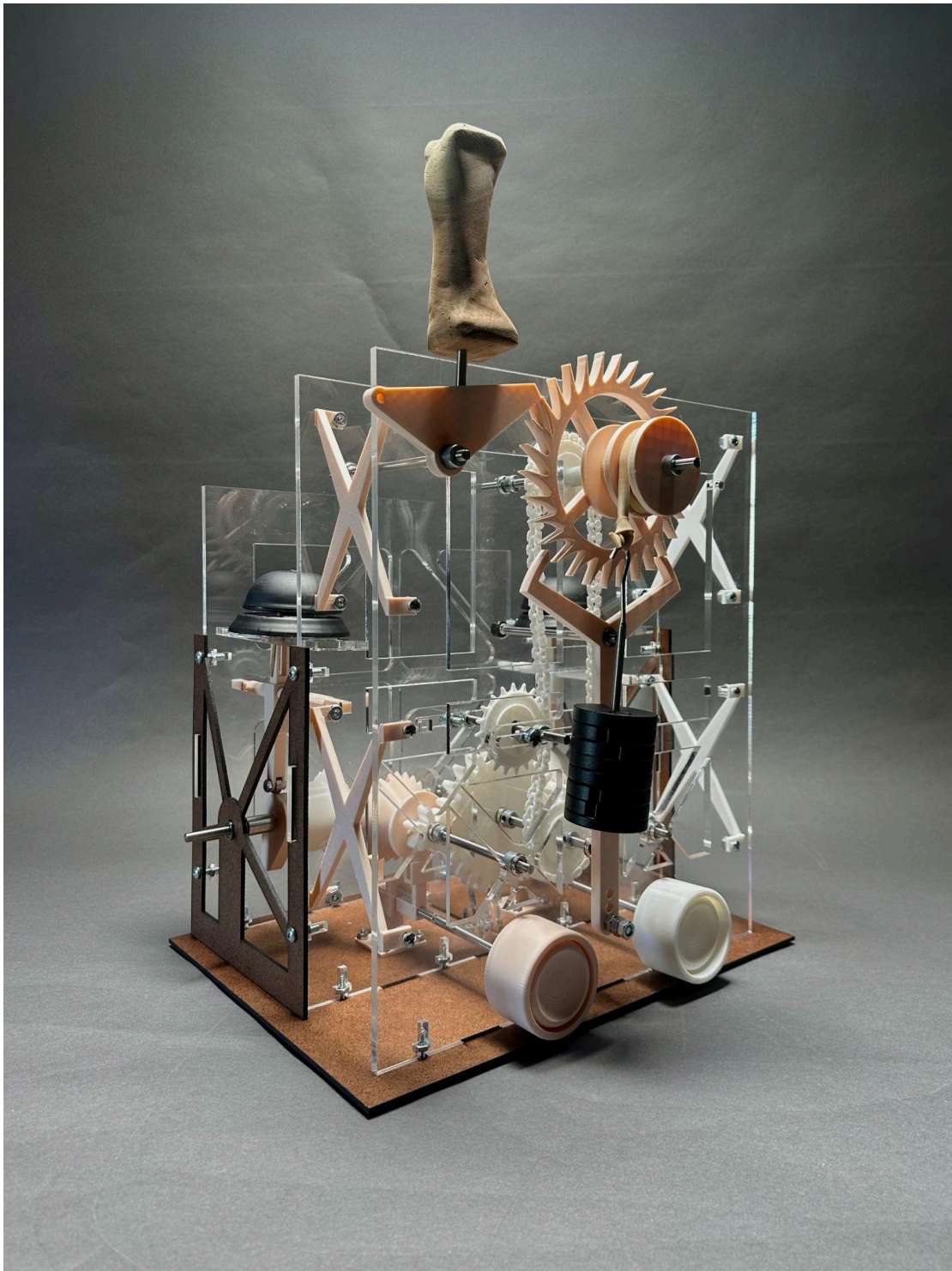


Figure 13: Second working prototype of the machine.

Based on insights from our force block diagram, in-class feedback, and our finite element analysis, we redesigned the machine to include two bells, clutch mechanisms to activate the bells, a wooden handle, and an optimized frame. This prototype was then polished and tweaked, resulting in the final prototype detailed in Figure 1.

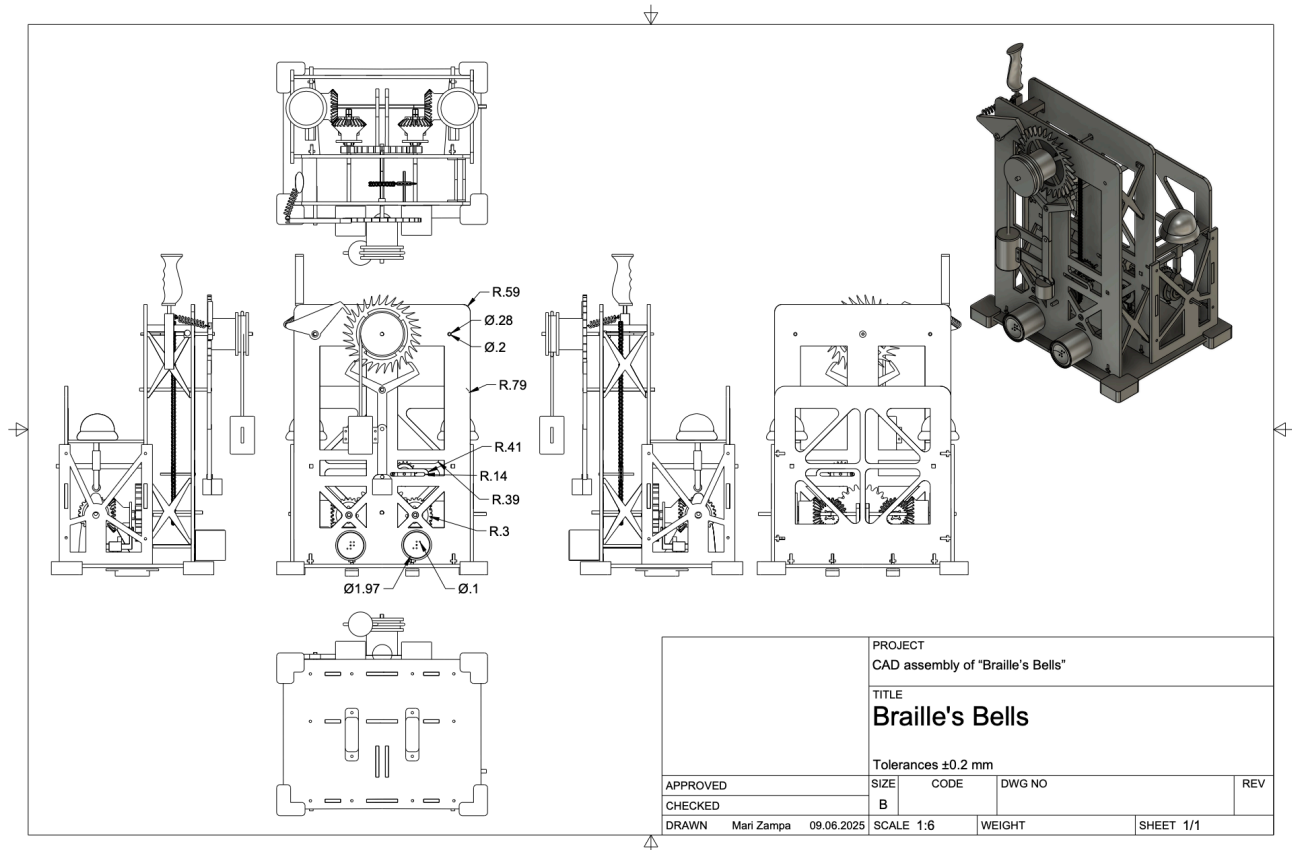


Figure 14: Final CAD assembly of "Braille's Bells"

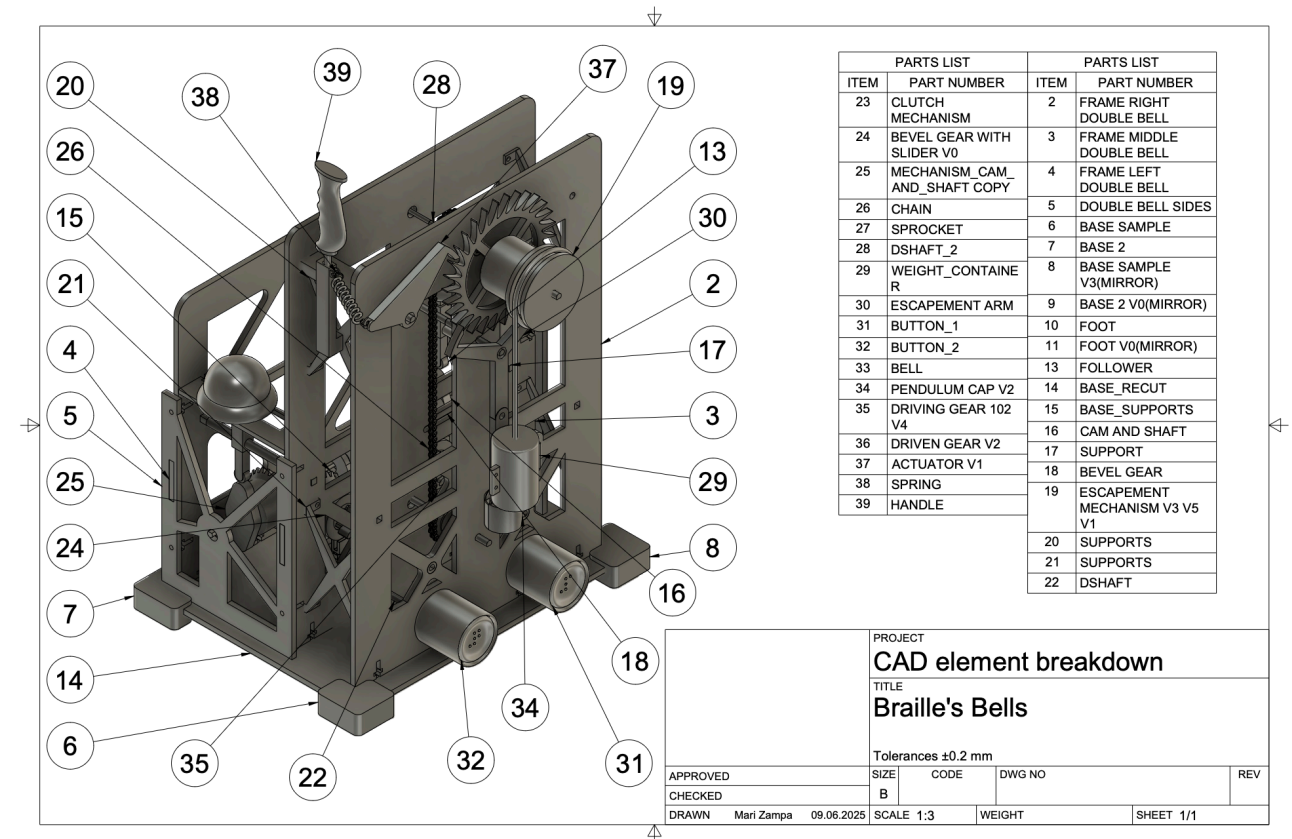


Figure 15: Final CAD element breakdown.